

MTH313M ASSIGNMENT 1

Instructions:

- (i) Each question is of 10 marks.
- (ii) Only one question will be evaluated. The question number (to be evaluated) will be declared after the deadline for submission.
- (iii) Deadline for submission: **10:00 am IST, Wednesday April 1, 2026.**
- (iv) **Assignment submission will be through the HelloIITK portal.** Email submissions and any late submission through the HelloIITK portal will NOT be graded.
- (v) Please **upload one single PDF file** containing all your answers.
- (vi) Provide all details/justifications.

- Question 1.
- (a) (4 marks) Let $S_n, n = 1, 2, 3, \dots$ denote the random times at which the n -th event occurs following a Poisson Process with rate $\lambda > 0$. Find the Conditional distribution of $(S_1, S_2, \dots, S_{n-1})$ given $S_n = t$.
 - (b) (2 marks) Let $\{N_t\}_{t \geq 0}$ be a Poisson Process with rate $\lambda > 0$. Compute $E[N_t N_{t+h}]$ for $t \geq 0, h > 0$.
 - (c) (4 marks) Let $W = \sum_{j=1}^N Y_j$ be a compound Poisson random variable with $N \sim \text{Poisson}(\lambda)$ for some $\lambda > 0$ and $Y_1, Y_2, \dots$ are discrete i.i.d. random variables with common p.m.f. f supported on the set of natural numbers $\{1, 2, \dots\}$. Show that W is also a discrete random variable supported on $\{0, 1, 2, \dots\}$ and express its p.m.f. in terms of λ and f .

- Question 2.
- (a) (5 marks) Let $\{X_t\}_{t \geq 0}$ and $\{Y_t\}_{t \geq 0}$ be independent Poisson Processes with rates $\lambda > 0$ and $\mu > 0$, respectively. Fix positive integers i and j . Consider the two-dimensional process $\{Z_t\}_{t \geq 0}$ with $Z_t = \begin{pmatrix} X_t \\ Y_t \end{pmatrix}, \forall t \geq 0$. Note that $Z_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. What is the probability that the process $\{Z_t\}_{t \geq 0}$ visits the point (i, j) ?
 - (b) (3 + 2 marks) Let $\{N_t\}_{t \geq 0}$ be a Conditional Poisson Process with rate λ being a absolutely continuous RV with the p.d.f.

$$f_\lambda(x) := \begin{cases} \frac{\alpha}{(\beta-1)!} e^{-\alpha x} (\alpha x)^{\beta-1}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

where β is a positive integer and $\alpha > 0$ is a real number. For non-negative integers n , compute $\mathbb{P}(N_t = n)$ and find the conditional distribution of λ given $N_t = n$.